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Applications of Passive Infrared Thermal Imaging for Detecting Delaminated Areas of Ancient Rock-cut Tunnel

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Abstract. Delaminations are generally developed in construction works due to numerous factors, including cyclic loading, water flow, and weather conditions. A conventional non-destructive method, such as tapping or chain dragging, is commonly used to predict subsurface delaminations. However, tapping tests may require more precision, especially for minor subsurface defects. Passive Infrared Thermal Imaging or Passive Infrared Thermography (PIRT) effectively detects sizable delaminated areas of a structure quickly. However, the inspected structural components are recommended to be exposed to direct sunlight. In this study, PIRT was used to investigate an operating ancient rock-cut tunnel and to detect potentially dangerous locations where delamination and spalling have occurred. The results prove that PIRT can effectively detect delaminated areas, even when there is only a 1-2°C atmospheric temperature difference nine hours before testing.

1. Introduction

Previously, road and mountain railway tunnels worldwide as well as in Vietnam were often not fully reinforced to reduce construction costs. The construction method usually involved blasting and excavating the material out of the tunnel, resulting in an uneven surface. Due to the lack of reinforcement, surface damage is common and poses a danger. Additionally, measurement and inspection are difficult. Figure 1 illustrates the surface of a blasted and unreinforced tunnel. Tunnels in operation face a potential danger of partial or complete collapse. Thus, a tunnel inspection program is always applied to carry out maintenance activities and/or rehabilitation of structures to ensure public safety. The inspection report is presented at an element-level, representing the condition of the elements, ranging from good to severe defects [1]. The inspection provides percentages of the total area for map cracking, moisture penetration, efflorescence, and delamination [2, 3].



Figure 1. Unlined rock-cut tunnel.



To conduct a comprehensive tunnel inspection, a combination of site observation, destructive and non-destructive methods are often utilized. Currently, the focus is on advanced non-destructive methods. For instance, according to the tunnel inspection guidelines of the Arizona Department of Transportation, non-destructive testing can complement the inspection process along with traditional methods like hammer sounding and destructive testing methods such as coring [2]. Other examples of the non-destructive inspections of the Tooth Rock and Vista Ridge Tunnels were conducted in 2018 in the R06G Tunnel Liner Investigation Project. The objective of the project was to determine the condition of the tunnel liner utilizing Ground Penetrating Radar (GPR), Passive Infrared Thermography (PIRT) techniques in combination with High Resolution Video Imaging (HRI). The results were that GPR was utilized to identify areas of delaminated and deteriorated concrete, voids behind the liner and areas of high moisture concentration within the liner and behind the liner, while the PIRT technique was used to locate areas of potential water flow and wet liner surfaces, debonded tiles, and delaminated concrete [4]. Konishi et al. [5] conducted several measurements and tests in our box-type tunnel and shield tunnel to confirm feasibility of applying the infrared thermometry for maintenance activities to detect the non-visible voids, then the results compared with those of other conventional test methods. The conclusion was cleared that voids in shield tunnel can be detected by the infrared thermometry. For cut-and-cover tunnel, the air temperature in the tunnel, concrete surface temperature, and air temperature of outside should be considered and the inspection of infrared thermography should be carried out in proper time.

The aim of the study is to conduct a PIRT in an ancient, unlined rock-cut tunnel. However, the rough inner surface of the tunnel poses a challenge in applying the PIRT method. Additionally, direct sunlight is lacking, and vehicles are still operating inside the tunnel. Despite these challenges, the study successfully predicts suspected delamination. The data presented in a visual map will be helpful for engineers preparing their repair plans and methods. The research provides inspection results, methodology, analysis, and discussion.

2. Methodology of Passive Infrared Thermal Imaging

The electromagnetic spectrum known as Infrared has a wavelength ranging from 0.8 μm to 1000 μm [6]. However, the PIRT method utilizes only a limited part of the IR spectrum, specifically from 0.8 μm to 14 μm , as most of the spectrum is obstructed by the atmosphere. The temperature of an object (T_{obj}) can be determined by using Equation 1, which relies on the total radiation energy (W_{tot}) detected by the infrared detector [7, 8, 9].

$$T_{obj} = \left[\frac{W_{tot} - (1 - \varepsilon) \cdot \tau \cdot \sigma \cdot T_{refl}^4 - (1 - \tau) \cdot \sigma \cdot T_{atm}^4}{\varepsilon \cdot \tau \cdot \sigma} \right]^{\frac{1}{4}} \quad (1)$$

Where W_{tot} is total radiation received by the IR camera, σ is the Stefan-Boltzmann constant ($=5.67 \times 10^{-8} \text{W/m}^2\text{K}^4$), and the other parameters must be set up in the camera including the emissivity of the object material (ε), the transmittance of the atmosphere (τ), the reflected temperature (T_{refl}), and the ambient temperature (T_{atm}). Thus, the thermal detector has been calibrated for emissivity of material, ambient and reflected temperature, distance, and relative humidity [10].

The underlying concept of PIRT is depicted in Figure 2. As solar radiation heats up the object under investigation, the temperature at the surface of the object will be higher than at its core. Consequently, heat flows through the object. However, when the heat flux reaches an obstacle boundary such as delamination or cracks, it bounces back to the surface, causing the defective regions to be warmer than the normal areas around them. Besides, the arrangement of the tunnel's surface is related to the rock structure, and indentations can improve the ability to emit heat radiation in a direction that is perpendicular to the surface, especially for highly smooth metal surfaces [11]. This rise happens because of both the direct and reflected radiation. Additionally, the indentations, hollows, and roughened surfaces of tunnels lead to uneven distribution of the heat radiation, which makes it challenging to distinguish between faulty and non-faulty regions. Furthermore, for the infrared

thermography technique to be effective in detecting delamination, the width of the delaminated area should be twice its depth [12, 13].

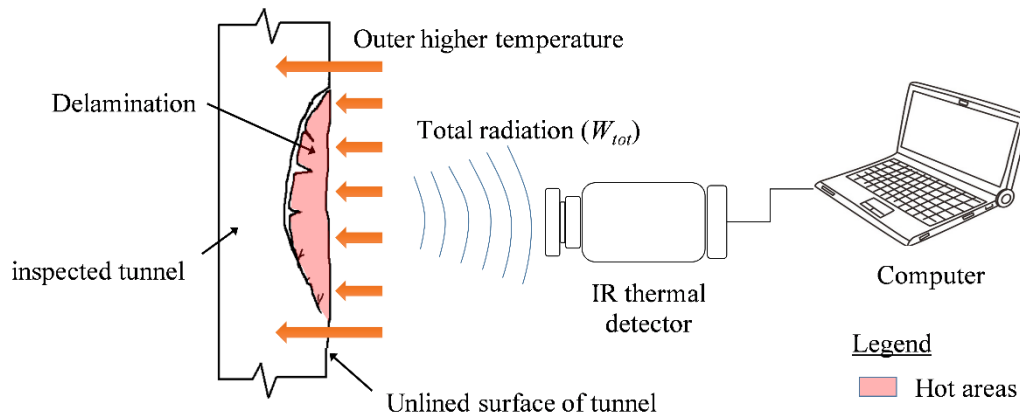


Figure 2. Thermal behavior of tunnel's surface detected by PIRT.

3. Tunnel inspection using PIRT

3.1. Infrared thermal detector setup

To locate the questionable areas, the IR thermal detector was initially used to scan the tunnel. Afterwards, it was positioned at a distance of approximately 3 meters perpendicular to the tunnel surface to capture the thermograms. Over 68 thermal images and 200 digital images were carefully captured at the suspected areas including sides and ceiling walls. The Field of view (FOV) is divided into three directions including the horizontal field of view (HFOV), the vertical field of view (VFOV), and the diagonal field of view (DFOV) as illustrated in Figure 3. They are calculated by the working distance ($L = 3\text{m}$, in this study) and the corresponding FOV angles α_H , α_V , and α_D as given by the following Equations 2, 3, 4 [14, 15]. In case of this study, the camera is placed at 3 m, and the α_H and α_V are 24°C and 18°C , respectively, thus the corresponding results of HFOV and VFOV are 1.28 m and 0.95 m.

The test captured all images based on the image plane (from HFOV and VFOV) to assess tunnel walls and ceiling, and then returned a map of deteriorations. The thermal detector used in this study was the FLIR SC660 with 640×480 pixels. Therefore, each pixel size is approximately 2 mm, which is acceptable for the purpose of the inspection, which is subsurface delamination detection.

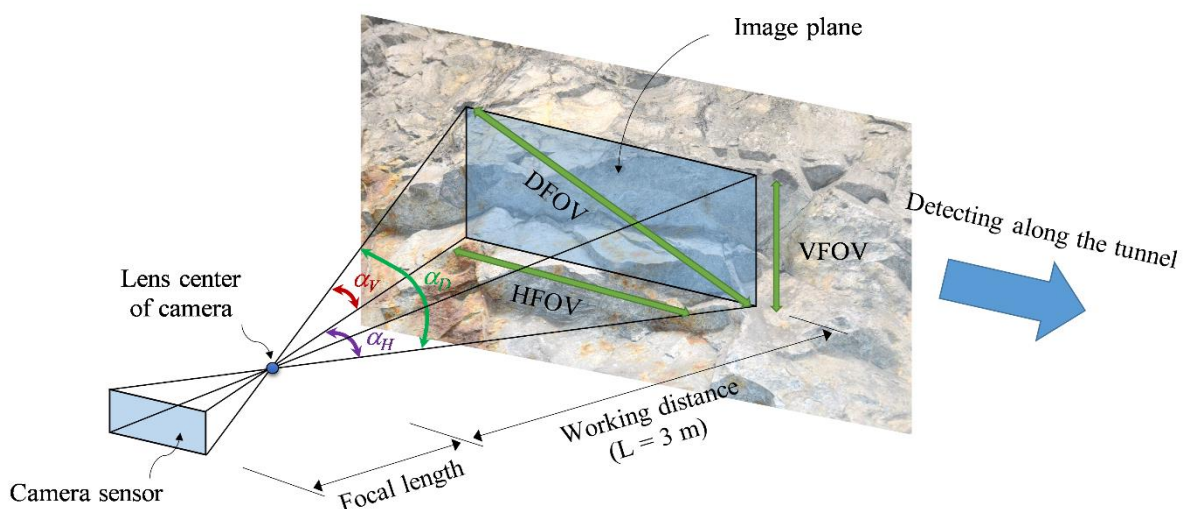


Figure 3. IR thermal detector set up at 3m from the tunnel surface.

HFOV, VFOV, and DFOV are calculated as follows:

$$\text{HFOV}(m) = 2L(m) \cdot \tan\left(\frac{\alpha_H(^{\circ}\text{C})}{2}\right) \quad (2)$$

$$\text{VFOV}(m) = 2L(m) \cdot \tan\left(\frac{\alpha_V(^{\circ}\text{C})}{2}\right) \quad (3)$$

$$\text{DFOV}(m) = 2L(m) \cdot \tan\left(\frac{\alpha_D(^{\circ}\text{C})}{2}\right) \quad (4)$$

3.2. Testing equipment

A long-wavelength high-resolution infrared thermal camera, FLIR SC660 was used to monitor and measure the variation in the temperatures of the tunnel. Some major technical data of the camera are listed in Table 1 [16]. The camera can automatically calculate the atmospheric transmission based on distance and relative humidity. It also provides tools to adjust the emissivity of material and to measure temperature at any types of Regions of Interest (ROI) such as spots, lines, and circle, rectangular, and polygon areas. Initial parameters used to set up the camera were summarized in Table 2.

Table 1. Technical data of FLIR camera SC660 [16].

Items	Parameters
IR resolution	640 x 480 pixels
Thermal sensitivity/NETD	<30 mK @ +30°C
Field of view (FOV)	24°x 18°/0.3m
Spatial resolution (IFOV)	0.65 mrad
Image frequency	30 Hz
Focal Plane Array (FPA)	Uncooled microbolometer
Wavelength	7.5 ÷ 13 μm
Temperature range	-40°C ÷ 120°C
Accuracy	±1°C or ±1% of reading

Table 2. Initial parameters for IR thermal detector.

Parameter	Values	Measuring Equipment
Emissivity	0.95	ACI [17]
Atmospheric temperature (°C)	17.5	CEM DT-615
Relative humidity (%)	75	CEM DT-615
Distance (m)	3.0	Laser Bosch GLM 50

3.3. Consideration of the thermal detector accuracy

The accuracy as indicated in Table 1 is the result of a widely used uncertainty analysis technique called “Root-Sum-of-Squares”. The overall error equation considers partial errors come from several variables (ΔT_i) including emissivity, reflected ambient temperature, transmittance, atmosphere temperature, camera response, and calibrator (blackbody) temperature accuracy. It can be written as follows [18],

$$\text{Total Error} = \sqrt{\Delta T_1^2 + \Delta T_2^2 + \Delta T_3^2 + \dots} \quad (5)$$

This means that any camera, in any environment condition (within specification), at any time will give a reading within the accuracy specification (e.g., $\pm 1^\circ\text{C}$). To calculate the accuracy specification, looking at the following example. Assuming that the surface temperature is 17.5°C , the FLIR SC660 has a specified accuracy of $\pm 1^\circ\text{C}$, hence the temperature reading of a blackbody can be 16.5°C to 18.5°C (emissivity ≈ 1.0); However, if the surface temperature is too high, ex. 175°C , the accuracy is not from $174 \sim 176^\circ\text{C}$ ($\pm 1^\circ\text{C}$). In this case, the reading could vary between 173.3°C and 176.8°C ($\pm 1\%$ of the reading). Besides, when measuring materials with emissivity less than one, e.g., 0.95, the measurement error is increased. A simple approach referring from Infrared Training Center (ITC) can be applied as Equation 6. Therefore, the adjusted accuracy from the above example would be $\pm 1^\circ\text{C}/0.95 = \pm 1.05^\circ\text{C}$ approximately.

$$\text{Adjusted accuracy} = \frac{\text{Camera accuracy specification}}{\text{surface emissivity}} \quad (6)$$

When we compare the temperatures of the defective and non-defective areas, we can see that the thermal sensitivity of the FLIR SC660 can detect differences as small as 0.03°C , as described in Table 1. As a result, the detector can provide accurate readings for tunnels that lack direct solar radiation.

4. Analysis and discussion

To avoid traffic congestion, the test was conducted at night when only a few bus routes were in operation. Inside the tunnel, only traffic signals and neon lights were available, so the digital camera should be set up with the night mode, and the images should be post-processed using image processing methods (if applicable). Additionally, the difference between the temperature inside and outside of the tunnel walls played a vital role in detecting subsurface defects. In this specific situation of the test, the temperature was approximately $10\text{--}11^\circ\text{C}$ from 0:00 to 6:00 am, with a significant change from 11 to 18°C from 6:00 am to 12:00 pm, and then reducing to around $16\text{--}17^\circ\text{C}$ from 12:00 pm to the testing time (around 9:00 pm). Thus, the temperature difference from noon to the testing time was only $1\text{--}2^\circ\text{C}$. As the temperature of the tunnel walls is close to the temperature of their surroundings, heat transfer is unlikely to occur, or at a very small rate, making it difficult to locate defects using PIRT. Thermal imaging results need to be compared with processed digital images to best locate the position of the defect.

The testing captured over 268 shots, including thermal and digital images, and the results indicate a few suspected defects in certain areas. Figure 4 shows the measurement results of a clear suspected defect location. Specifically, the results indicate that the rock mass in Figure 4(a) may have experienced internal delamination. This finding is further confirmed by the thermal image results in Figure 4(b). When analyzing the surface temperature of the tunnel wall through the cross-section A-A, the temperature results (calculated using Equation 1) show that the section from the pixel 50 to 200 in Figure 5 produces a temperature result of approximately 15°C . Meanwhile, the surface temperature of the surrounding rock is only about 14.6°C , with a difference of 0.4°C . As mentioned in Section 3.3, the device's thermal sensitivity can detect differences up to 0.03°C . Therefore, this measurement result is highly reliable. The findings demonstrate that PIRT can accurately identify sections where delamination has occurred, even when there is no direct exposure to sunlight. To determine the size of the detached rock using the IR detector setup, we measured the horizontal dimension of the rock block to be around 150 pixels, which is approximately 30 cm.

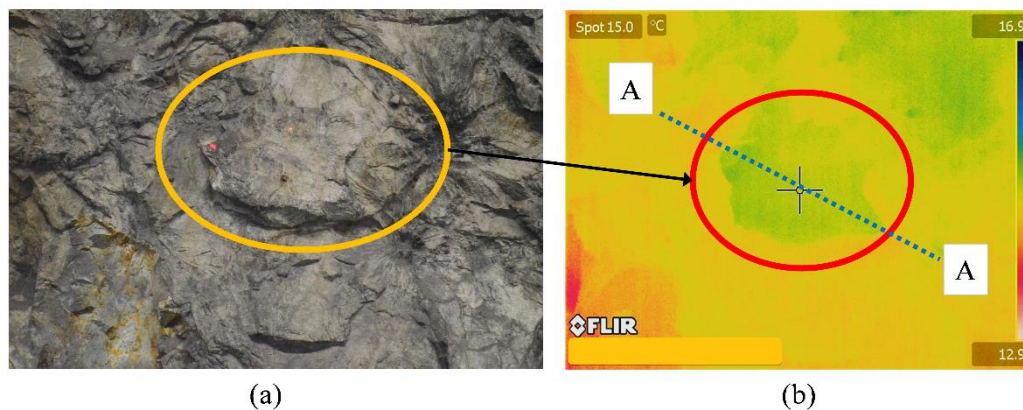


Figure 4. Subsurface delaminated areas detected by PIRT.

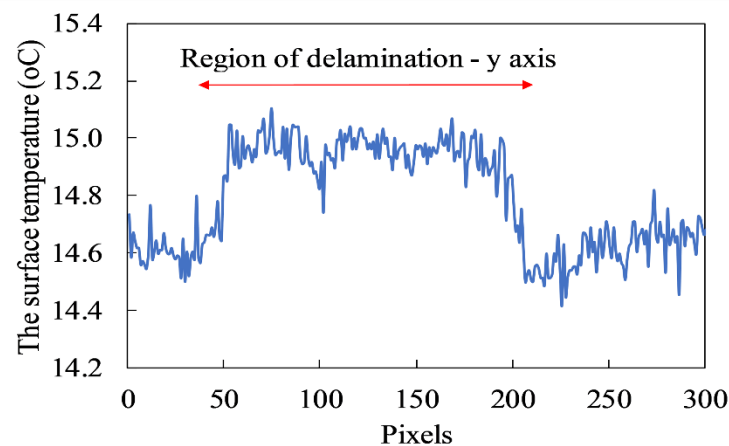


Figure 5. Analysis of surface temperature (cross section A-A).

5. Conclusions

The study focuses on the detectability of PIRT for an unlined rock-cut tunnel where there is no direct solar energy. The test was conducted using a high-resolution thermal detector with an uncooled microbolometer focal plane array of 640 x 480, which allows for a thermal sensitivity of 0.03°C, resulting in highly reliable results. The results prove that PIRT can effectively detect delaminated areas, even when there is only a 1-2°C atmospheric temperature difference nine hours before testing, indicating a low thermal contrast between defective and non-defective areas. Additionally, the combination of PIRT with digital images is important as it allows for the distinction between delaminated areas and other weathered rock positions. It is clearly seen that PIRT is capable of detecting potential delaminated areas in an unlined tunnel, even when the change in atmospheric temperature is not significant.

Acknowledgment

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References

- [1] Massachusetts Department of Transportation 2018 *Tunnel Inspection Handbook*
- [2] Arizona Department of Transportation-Bridge Group 2021 *Tunnel Inspection Guidelines* Arizona Department of Transportation
- [3] Tran QH, Dang QM, Pham XT, Truong TC and Nguyen TX 2023 Passive infrared

- thermography technique for concrete structures health investigation: case studies *Asian Journal of Civil Engineering* **24** 1323-1331
- [4] Nako A 2018 *Final Report on the Non-Destructive Inspection of the Tooth Rock & Vista Ridge Tunnels* NY 14304
 - [5] Konishi S, Kawakami K and Taguchi M 2016 Inspection method with infrared thermometry for detect void in subway tunnel lining *Procedia Engineering* **165** 474-483
 - [6] Usamentiaga R and et. al. 2014 Infrared Thermography for Temperature measurement and non-Destructive Testing *Sensors* **14** 12305-12348
 - [7] Maldague XP 2001 *Theory and Practice of Infrared Technology for Nondestructive Testing* John Wiley & Sons NY
 - [8] Huh J, Tran QH, Lee J, Han D and Yim S 2016 Experimental Study on Detection of Deterioration in Concrete Using Infrared Thermography Technique *Advances in Materials Science and Engineering Article ID 1053856* 12
 - [9] Tran QH, Han D, Kang C, Haldar A and Huh J 2017 Effects of Ambient Temperature and Relative Humidity on Subsurface Defect Detection in Concrete Structures by Active Thermal Imaging *Sensors* **17(8)** 1718
 - [10] Tran QH, Huh J, Mac VH, Kang C and Han D 2018 Effects of rebars on the detectability of subsurface defects in concrete bridges using square pulse thermography *NDT and E International* **100** 92-100
 - [11] Vollmer M and Mollmann KP 2010 *Infrared Thermal Imaging - Fundamentals, Research and Applications* Viley-VCH Weinheim Germany
 - [12] Tran QH, Huh J, Kang C, Lee BY, Kim IT and Ahn JH 2018 Detectability of Subsurface Defects with Different Width-to-Depth Ratios in Concrete Structures Using Pulsed Thermography *Journal of Nondestructive Evaluation* **37(32)** 1-10
 - [13] Tran QH 2021 Passive and Active Infrared Thermography Techniques in Nondestructive Evaluation for Concrete Bridge. *Proc. of the 1st International Conference on Innovations for Computing Engineering and Materials* HCMC Vietnam
 - [14] Holst GC 2000 *Common Sense Approach to Thermal Imaging* Spie Optical Engineering Press, Bellingham Washington
 - [15] Maldague XP 1993 *Nondestructive Evaluation of Materials by Infrared Thermography* Springer-Verlag
 - [16] FLIR SC660 Catalog 2014 *Technical Data of FLIR SC660 Infrared Camera* FLIR System Inc.
 - [17] ACI 2013 *Report on Nondestructive Test Methods for Evaluation of Concrete in Structures* American Concrete Institute Farmington MI
 - [18] FLIR 2016 Infrared Camera Accuracy and Uncertainty in Plain Language Available at: <http://www.flir.com/science/blog/details/?ID=74935>